Finding the best confidence experiment

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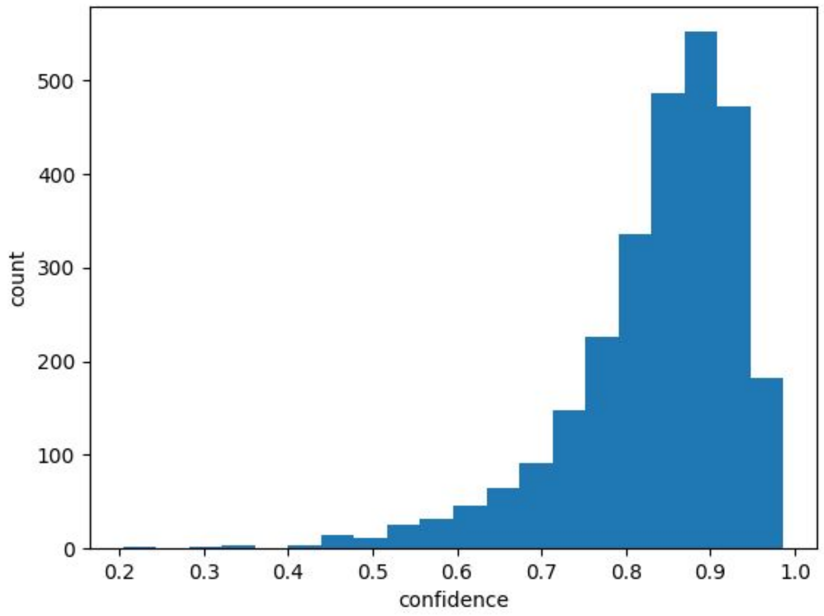
First step

We obtained the results, along with their associated confidence scores, from the last experiment, indicating that the prompt does not impose a condition on confidence. Now, we regard these results as data for identifying the best confidence experiment.

Here is the one example of our data for this experiment


```
{
  "asr_transcription": {
    "text": " On the I'll do it more completely.",
    "confidence_score": 0.79
  },
  "reference_transcription": "ONLY I'LL DO IT MORE COMPLETELY",
  "corrected_asr_transcription": "I'll do it more completely."
}
```

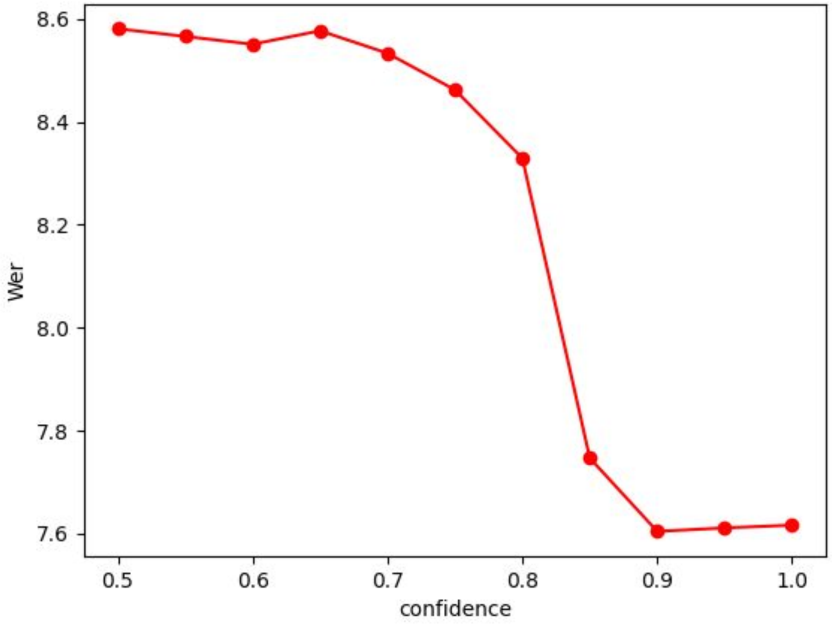
Histogram plot of the confidence values of the Librispeech transcriptions



Based on the histogram on last slide, Enno suggested me to change the range of threshold.

Threshold = [0.5, 0.55, 0.6, 0.65, ..., 1.0]

Wer versus confidence plot



Next step

After determining a confidence threshold, we can incorporate this threshold as a condition in the prompt for the next experiment: 'Experiment with the best confidence.

Here is the example of prompt for next experiment

```
{       'role': 'system',
        'content': f"""You are a helpful assistant that corrects ASR errors. \
        You will be provided with the ASR output in JSON format with keys: text and confidence_score. \
        The text is the ASR transcription for an audio and confidence_score is its level of confidence. \
        If the confidence_score is low (lower than 0.7), it's very likely that the ASR system made an error in the transcription. \
        Therefore, your task is to replace low-confidence text in the ASR transcription with better transcription that make sense in the context. \
        Provide the most the corrected transcription in string format. \
        Do not change the case, for example, lower case or upper case, in the transcription. \
        Do not output any additional text that is not the corrected transcription. \
        Do not write any explanatory text that is not the corrected transcription.
        """
}
```

My team’s suggestions about this experiment

Mathew suggested me to use word-level confidence score instead of sentence-level confidence score.

Enno suggested me to start by just taking the worst of the word-level values and use that as a new "sentence-value" confidence.

Mathew and Alexandre suggested me to use GPT-4 on experiment without confidence to compare the new results with the results that I have already from GPT-3.5 turbo.

Thank you!